

Mingu Kang

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Research Interests

I am an M.S./Ph.D. student at the Visual Information Processing Lab, Seoul National University, advised by Prof. Joonseok Lee. My research centers on video understanding and world models. I study multimodal representations that capture temporal dynamics, high-level semantics, and physical interactions, with applications to long-form video understanding, summarization, retrieval, motion understanding, and planning.

Education

Seoul National University, Seoul, Republic of Korea Mar 2026 – Present
M.S./Ph.D. Student

Sungkyunkwan University, Suwon, Republic of Korea Mar 2022 – Feb 2026
B.S. in Applied Artificial Intelligence
◦ GPA: 4.29/4.5 (Overall), 4.46/4.5 (Major); graduated *magna cum laude*.

Publications

Anisotropic Representations Improve Planning in JEPA World Models 2026

Mingu Kang, Yoori Oh[†], Soo Kyung Kim[†], Joonseok Lee[†]
Preprint

DRUM: Dense Reliability Utilization Module for Robust Autonomous Driving Perception 2026

Junhyung Kim, *Mingu Kang*, YunSeok Choi
Preprint

Do Not Stop at Frame Scoring: Enabling End-to-end Training for Video Summarization 2026

Sumin Kim*, *Mingu Kang**, Minjun Kim, Sungmin Yang, U Kang, Jaegul Choo[†], Yoori Oh[†], Joonseok Lee[†]
Preprint

From Matching to Reasoning: Query-Aware Long Video Summarization 2026

Mingu Kang, Sumin Kim, Hyunjin Lee, Sungmin Yang, Yoori Oh[†], Joonseok Lee[†]
Conference on Neural Information Processing Systems (NeurIPS)

TripleSumm: Adaptive Triple-Modality Fusion for Video Summarization [🔗](#) 2026

Sumin Kim*, Hyemin Jung*, *Mingu Kang**, Yejin Kim, Yoori Oh[†], Joonseok Lee[†]
International Conference on Learning Representations (ICLR)

DisCoRD: Continuous Motion via Rectified Flow Decoding [🔗](#) 2025

Jungbin Cho*, Junwan Kim*, Jisoo Kim, Minseo Kim, *Mingu Kang*, Seuneun Hong[†], Tae-Hyun Oh[†], Youngjae Yu[†]
International Conference on Computer Vision (ICCV), Highlight

*Equal contribution. [†]Corresponding author.

Patents

Video Summarization Method and Computer Apparatus for the Same 2026

Mingu Kang, Sumin Kim, Hyemin Jeong, Yejin Kim, Yoori Oh, Joonseok Lee
Korean patent application filed May 2026; application number pending.

Experience

Seoul National University, Seoul, Republic of Korea May 2025 – Mar 2026
Full-time Research Intern (Advisor: Joonseok Lee)

Visual Information Processing Lab [🔗](#)

- Conducted research on multimodal video summarization and long-form video understanding.

Graduate School of AI at KAIST, Seoul, Republic of Korea Aug 2024 – Oct 2024

Full-time Research Intern (Advisor: Beomjoon Kim)

Humanoid Generalization Lab [🔗](#)

- Conducted research on robot planning using 3D representations and large language models.

Sungkyunkwan University, Suwon, Republic of Korea Dec 2023 – Jul 2024

Part-time Research Intern (Advisor: Yusung Kim)

Computer System Intelligence Lab [🔗](#)

- Conducted research on robot foundation models using diffusion models.

Projects

Building Foundation Models and Evaluation Benchmarks for Video Summarization

NRf Basic Science Research Program, Sep 2026 – Present

Leading research on multimodal foundation models and evaluation benchmarks for video summarization under the 2026 program.

Honors & Awards

LG AI Research President's Award [🔗](#)

Sep 2026

LG AI Research

- Developed an AI model to predict pitch-control success probabilities.

Presidents' Awards, Korea East-West Power and GS E&R [🔗](#)

Sep 2026

3rd Wind Power Generation Forecasting AI Competition (BARAM 2026)

- Developed an AI model for wind power generation forecasting.

Best Research Award

Nov 2025

SKKU IME Annual Research Review

- Top-rated research selected by industry and government-funded institute evaluators.

Excellence Award (Korea Energy Agency) [🔗](#)

Sep 2025

Power Consumption Forecasting AI Competition

- Developed a power consumption forecasting AI model.

President's Award, Institute for Information & Communications Technology

Aug 2025

Planning & Evaluation (IITP) [🔗](#)

Institute for Information & Communications Technology Planning & Evaluation

- Developed paragraph-level classifiers for reliable detection of human-written and AI-generated text.

Intelligent Multimedia Research Center Director's Award

Jun 2025

Sungkyunkwan University

- Developed prompting and retrieval-augmented generation strategies for fair, neutral responses.

Minister of Employment and Labor Award

Apr 2025

LG AI Research

- Developed an AI model to predict the likelihood of pregnancy success in infertility treatment.

Sungkyunkwan University brAIn Research Program Award

Feb 2025

Sungkyunkwan University

- Conducted research on CoReaP, an image inpainting model using collaborative reconstruction with assistive priors.

President's List, High Student Success Performance

Jan 2025

Sungkyunkwan University

- Selected as one of the top 20 students for outstanding achievement across the university.

Korea Water Resources Corporation President Award

Dec 2024

K-water

- Developed an AI model to detect abnormal timing and leakage sections in water supply networks.

Korea Agro-Fisheries Food Trade Corporation Award

Nov 2024

Ministry of Agriculture, Food and Rural Affairs

- Developed an AI model for agricultural price forecasting.

Korea Computer Education President Award

Oct 2024

The Journal of Korean Association of Computer Education

- Developed a text-to-3D motion generation model and rehabilitation exercise web application.

Hankuk University of Foreign Studies Presidential Award

Oct 2024

Hankuk University of Foreign Studies

- Developed an AI model that classifies news categories from titles and text keywords.

SKKU University AI Director Award

Sep 2024

Sungkyunkwan University

- Developed an AI model to detect illegal gambling advertisements on web pages.

National Institute of Korean Language Award [🔗](#)

Aug 2024

National Institute of the Korean Language

- Developed an AI model to distinguish conversational context in Korean large language models.

Minister of Science and ICT Award [🔗](#)

Aug 2024

SW-Centered University

- Developed an AI model to detect AI-generated fake voices.

Dean's List Award, High Scholastic Achievement*Jul 2024**Sungkyunkwan University*

- Ranked in the top 1% for the Fall 2023 semester.

Fellowships & Scholarships

Master's Excellence Scholarship (Science and Engineering)*Mar 2026**Korea Student Aid Foundation***NRF Graduate Research Encouragement Grant***Sep 2026**National Research Foundation of Korea***Presidential Science Scholarship***Feb 2024**President of the Republic of Korea***Academic Services**

Reviewer*2026**Conference on Neural Information Processing Systems (NeurIPS)*